

Software Engineer — Frontend & Machine Learning / AI

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PROFILE

Software Engineer with over six years of professional experience — including three years delivering production frontend systems across a multi-brand estate at a FTSE40 company — and a BSc (Hons) in Machine Learning & AI graduating 2026. The combination of deep frontend expertise and hands-on ML/AI knowledge positions me for roles building intelligent, AI-powered products where both skill sets are used together — most recently a production-shaped RAG pipeline (FastAPI, local embeddings, vector search, grounded LLM answers with retrieval guardrails and an evaluation harness) and an end-to-end deep learning pipeline for medical imaging. Early adopter of AI-assisted development (GitHub Copilot in production since 2024; working with AI models since 2022). Based in Cape Town, UTC+2 — fully aligned with UK and EU working hours. Open to remote Senior or Intermediate Software Engineering roles, and ML/AI engineering roles at companies building intelligent products.

FEATURED PROJECTS

Deep Learning Breast Cancer Classification — CM3070 Final Project, University of London (2026)github.com/peter-kautwima/DeepLearning-BreastCancer-Classification · Final project grade: 74%

- Built an end-to-end deep learning pipeline for malignant/benign mammogram classification on the CBIS-DDSM clinical dataset (~3,500 biopsy-verified ROI patches).
- Four-stage iterative pipeline: Baseline CNN → Frozen VGG16 → Regularisation (Dropout + L2) → Fine-tuning. Best model: **AUC 0.7678, sensitivity 0.7055** on 714 held-out images — 64% relative sensitivity gain over baseline, exceeding mean individual radiologist AUC (McKinney et al. 2020).
- Implemented Grad-CAM explainability via a custom GradientTape workaround for a Keras 3 graph disconnection issue — heatmaps for all four prediction categories (TP, TN, FP, FN).
- Fully reproducible on a public dataset with stratified splits, class-imbalance weighting, and training-only augmentation.

Python · TensorFlow/Keras · Jupyter Notebooks · NumPy · Pandas · Scikit-learn · Matplotlib · pydicom · Google Colab (T4 / A100 GPU)

Document Q&A Service — Retrieval-Augmented Generation (RAG) Pipelinegithub.com/peter-kautwima/clickatell-assessment · Technical assessment, Clickatell — Innovation R&D Engineer (Intermediate) · 2026

- Built an end-to-end **RAG pipeline**: document ingestion and chunking → local embedding generation → vector similarity search → grounded LLM answers returned with their source chunks and relevance scores.
- FastAPI backend in a layered service architecture (chunking, embedding, retrieval, answering) behind thin HTTP routes, with Pydantic schemas as the single request/response contract.
- Embeddings generated locally with sentence-transformers (all-MiniLM-L6-v2), warmed at startup — no external embedding dependency. **Vector store implemented directly on NumPy** behind a VectorStore interface, so the in-memory implementation can be swapped for a hosted store without touching callers.
- Anthropic API integration for answer generation, with a keyless mock exercising the real prompt template so the service runs end to end without credentials.
- Hallucination guardrail**: a similarity threshold on retrieval returns "no relevant content" for off-topic questions instead of a fabricated answer.
- 61 tests at 99% statement coverage**, plus an opt-in **evaluation harness** grading retrieval quality against known Q&A pairs — asserting the fact-bearing chunk ranks top-5 above the relevance floor, and that the guardrail fires on out-of-scope questions.
- React + TypeScript (Vite) frontend with a typed API client mirroring the backend schemas; documented architecture decision record covering chunking strategy, embedding choice, vector storage, prompt design, concurrency, scaling to 1,000 documents / 100 concurrent users, AWS deployment and safety guardrails.

Python · FastAPI · sentence-transformers · NumPy · Anthropic API · RAG · Vector search · Embeddings · Prompt engineering · pytest · React · TypeScript · Vite

EXPERIENCE

Intermediate Software Engineer — The Rank Group (ShiftTech)

Dec 2022 – Feb 2026

FTSE40 international gaming & entertainment · 7,700+ employees globally · Cape Town subsidiary: ShiftTech

- Delivered frontend across a multi-brand unification initiative consolidating the full brand estate — including Mecca Bingo and Grosvenor Casino — onto a single Umbraco CMS, reducing per-brand maintenance overhead and enabling consistent feature rollout across a core engineering team of 10.
- Built a real-time jackpot carousel aggregating live data from multiple providers. Designed a Cloudflare Worker to poll and cache jackpot values at near-real-time frequency; built a Remix service layer to push live updates to the UI with smooth animations. Solved multi-provider data normalisation, polling architecture, and cache invalidation under production traffic. Live on Mecca Bingo and Grosvenor Casino.
- Delivered a redesigned registration flow across multiple brands, improving conversion and onboarding UX.
- Built and maintained a shared Storybook component library used by the 8-person frontend team, standardising UI components across the full brand estate.

TypeScript · React · Remix · GraphQL · Tailwind CSS · Storybook · Umbraco CMS · Cloudflare Workers · Git

Frontend Developer — Bertha Centre, UCT Graduate School of Business

Jul 2022 – Mar 2023

Part-time retainer · Leading social innovation research hub

- Maintained and extended the public-facing web presence for the Bertha Centre.

JavaScript · Node.js · GraphQL · GatsbyJS · Tailwind CSS · Netlify CMS · Netlify

Freelance Frontend Developer — LaTierra

2022

Freelance · Professional website

- Built the frontend for a professional website using a modern full-stack TypeScript setup.

Next.js · Prisma · Tailwind CSS · tRPC

Frontend Developer — reOS

Jul 2021 – Oct 2022

PropTech startup · Property management platform · Joined as intern, promoted to full-time Junior after 4 months

- Built production features across the rental management platform: payment reconciliation tool, user invitation & role permission workflows, file management, and dashboard statistics.
- Bridged development and customer-facing operations teams; took on sprint PM functions and led QA follow-up on critical issues.

TypeScript · React · Redux · RxJS · JavaScript · HTML · CSS/Sass · RESTful API

Freelance WordPress Developer

2019

- Built tailored websites for small businesses across branding, UX, and functionality. Managed full project lifecycle independently from client brief through to delivery.

Content Creator — Pocket Jam

2019

EdTech mobile learning application

- Created educational learning content for the Pocket Jam mobile app.

Founder — Supa Tutor & Mathamagicians (PTY)

May 2011 – Jun 2021

EdTech startup · Maths, English & Coding · 10 years

- Founded and ran an EdTech tutoring platform for 10 years. Began building the interactive online learning platform in 2017 and successfully piloted it at Summit House College in 2018. Developed and delivered a coding curriculum to students. Received an acquisition offer in 2015; chose to continue building.

Substitute Teacher — Summit House School

Mar 2018 – Dec 2018

Private school · Hout Bay, Cape Town

- Substitute teacher for Grade 8–12 in English, Mathematics and Science. Concurrently trialled the SupaTutor online Maths course with Matric students at the school.

Financial Director & Head Booker — The Fantastic Agency

Nov 2019 – Feb 2021

- Managed a portfolio of 250 models; oversaw financials, bank reconciliation, and booking operations. Revenue increased twofold over two years.

TECHNICAL SKILLS

FRONTEND (PROFICIENT)

TypeScript · JavaScript · React · Redux · GraphQL · HTML · CSS/Sass · Tailwind CSS · Storybook · CSS Modules · Remix · GatsbyJS · RxJS

INFRASTRUCTURE & TOOLING

Cloudflare Workers · Netlify · Umbraco CMS · Git / GitHub · RESTful APIs

AI TOOLING

GitHub Copilot (production, 2+ yrs) · Claude Code · LLMs & AI models since 2022 · Prompt engineering · Prompt templates & evaluation

MACHINE LEARNING / AI

Python · TensorFlow/Keras · Deep Learning · RAG (Retrieval-Augmented Generation) · Vector search & embeddings · sentence-transformers · LLM integration (Anthropic API) · LLM evaluation harnesses · Hallucination guardrails · FastAPI · Transfer Learning · CNNs · Neural Networks · Computer Vision · Grad-CAM · Data Science · Genetic Algorithms · Evolutionary Algorithms · NumPy · Pandas · Scikit-learn · Matplotlib · pydicom · Jupyter Notebooks · Google Colab

ALSO USED

Node.js · SQL · Express.js · Next.js · Prisma · tRPC · Bootstrap · Jest · Cypress · C++ · Docker

EDUCATION

BSc (Honours) Computer Science — Machine Learning & Artificial Intelligence	2020–2026
University of London (Goldsmiths) · Graduating 2026	
Notable modules: Artificial Intelligence · Machine Learning · Deep Learning · Natural Language Processing · Intelligent Signal Processing · Graphics Programming · Computer Security · Data Structures & Algorithms · Computational Mathematics	
Shaw Academy — Web Development, Web Design, Financial Trading & Investment	2019
Google IT Support Professional Certificate	2022
University of Cape Town — 2nd year, Chartered Accounting & Information Systems	2009–2010
Camps Bay High School — Matriculated, 7th in grade	2008
Deputy Head Prefect · Head of Debating · 1st Team Tennis · 1st Team Squash · 2nd Team Rugby	



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To Whom It May Concern,

I am pleased to write this reference letter for Peter Kautwima-Peetz, who worked at The Rank Group as a Software Engineer from December 2022 to February 2026.

He contributed meaningfully to various projects, most notably the full platform re-build during his tenure at ShiftTech. Peter demonstrated a strong ability to solve complex problems and collaborate effectively with colleagues.

Peter's work ethic and attention to detail made him a strong member of our team. He was always eager to learn new technologies and took initiative in improving our software solutions. His positive attitude and reliability ensured that tasks were completed on time.

I am confident that Peter will be a great asset wherever he chooses to apply his talents. Please feel free to contact me if you require further information.

A handwritten signature in black ink, appearing to read "Chris Marais", with a long horizontal flourish extending to the right.

Sincerely,
Christopher Marais
Chief Engineer
The Rank Group



Reference Letter

PETER KAUTWIMA

26 November 2022

To whom it may concern,

I can confirm that Peter Kautwima completed a four month internship with reOS from 11 July 2021 to 10 November 2021. Upon the successful completion of his internship he was appointed as a full time Junior Front-End Software Developer. Peter was a full time reOS employee, in the mentioned designation, from 11 November 2021 to 3 October 2022.

During his time with reOS, Peter worked on the following noteworthy projects, amongst other small enhancements and bug fixes:

1. An internal admin tool for assisting the support team in reconciling and allocating payments with incorrect references. The wallet service auto-allocates payments with correct references, but there are occurrences where payments are made with an incorrect reference and these need to be manually allocated accurately and efficiently. This tool enabled the support team to view and search for payments with incorrect references easily, and provide supporting information to assist the decision making process. Once a payment was allocated, the tool would execute the payment instruction.
2. File management and dashboard stats.
3. Optimising our sorting and filtering throughout the UI of the web app.
4. The UI implementation for an integration with a business partner of reOS, that provides adjacent services relating to property maintenance services.
5. The extension of our user invitation management workflows and user role permission settings, while improving the overall UX.

Over and above his technical function, Peter was a great bridge between the development team and the customer facing operations team. He also fulfilled some small project management functions within our sprints and played a very active role in following up on critical systems issues and QA tasks.

Unfortunately, due to a restructure in October 2022, Peter's role was made redundant and he was retrenched on 3 October 2022.

Yours sincerely

A handwritten signature in black ink, appearing to read "Craig Buckley". The signature is fluid and cursive, with a large initial "C" and "B".

Craig Buckley
MANAGING DIRECTOR